

DSA Topics, Patterns, and LeetCode Questions

1. Arrays

Subtopics

- Array traversal
- Two pointers
- Sliding window
- Prefix sums
- Kadane's Algorithm
- In-place modification
- Matrix traversal
- Intervals

LeetCode

- Two Sum
<https://leetcode.com/problems/two-sum/>
- Best Time to Buy and Sell Stock
<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/>
- Product of Array Except Self
<https://leetcode.com/problems/product-of-array-except-self/>
- Maximum Subarray
<https://leetcode.com/problems/maximum-subarray/>
- 3Sum
<https://leetcode.com/problems/3sum/>

- Container With Most Water

<https://leetcode.com/problems/container-with-most-water/>

- Subarray Sum Equals K

<https://leetcode.com/problems/subarray-sum-equals-k/>

2. Strings

Subtopics

- Character frequency counting
- Two pointers
- Sliding window
- String parsing
- Palindromes
- Anagrams
- Substrings

LeetCode

- Valid Anagram

<https://leetcode.com/problems/valid-anagram/>

- Valid Palindrome

<https://leetcode.com/problems/valid-palindrome/>

- Longest Substring Without Repeating Characters

<https://leetcode.com/problems/longest-substring-without-repeating-characters/>

- Longest Repeating Character Replacement

<https://leetcode.com/problems/longest-repeating-character-replacement/>

- Group Anagrams

<https://leetcode.com/problems/group-anagrams/>

- Longest Palindromic Substring

<https://leetcode.com/problems/longest-palindromic-substring/>

3. Linked Lists

Subtopics

- Traversal
- Slow and fast pointers
- Cycle detection
- Reversing a linked list
- Merging linked lists
- Finding the middle
- Dummy node technique
- Pointer manipulation

LeetCode

- Reverse Linked List
<https://leetcode.com/problems/reverse-linked-list/>
- Linked List Cycle
<https://leetcode.com/problems/linked-list-cycle/>
- Middle of the Linked List
<https://leetcode.com/problems/middle-of-the-linked-list/>
- Merge Two Sorted Lists
<https://leetcode.com/problems/merge-two-sorted-lists/>
- Remove Nth Node From End of List
<https://leetcode.com/problems/remove-nth-node-from-end-of-list/>
- Reorder List
<https://leetcode.com/problems/reorder-list/>

4. Trees and Tries

Subtopics

- DFS
- BFS / level-order traversal
- Preorder traversal
- Inorder traversal
- Postorder traversal
- Binary Search Trees
- Tree height and depth
- Lowest Common Ancestor
- Tree construction
- Trie insertion
- Prefix search
- Word search

LeetCode

- Maximum Depth of Binary Tree
<https://leetcode.com/problems/maximum-depth-of-binary-tree/>
- Invert Binary Tree
<https://leetcode.com/problems/invert-binary-tree/>
- Binary Tree Level Order Traversal
<https://leetcode.com/problems/binary-tree-level-order-traversal/>
- Validate Binary Search Tree
<https://leetcode.com/problems/validate-binary-search-tree/>
- Lowest Common Ancestor of a Binary Search Tree

<https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/>

- Diameter of Binary Tree

<https://leetcode.com/problems/diameter-of-binary-tree/>

- Implement Trie

<https://leetcode.com/problems/implement-trie-prefix-tree/>

- Word Search II

<https://leetcode.com/problems/word-search-ii/>

5. Stacks and Queues

Subtopics

- Stack operations
- Queue operations
- Monotonic stack
- Parentheses matching
- Expression evaluation
- BFS queues
- Min/max tracking

LeetCode

- Valid Parentheses

<https://leetcode.com/problems/valid-parentheses/>

- Min Stack

<https://leetcode.com/problems/min-stack/>

- Evaluate Reverse Polish Notation

<https://leetcode.com/problems/evaluate-reverse-polish-notation/>

- Daily Temperatures

<https://leetcode.com/problems/daily-temperatures/>

- Largest Rectangle in Histogram

<https://leetcode.com/problems/largest-rectangle-in-histogram/>

- Implement Queue Using Stacks

<https://leetcode.com/problems/implement-queue-using-stacks/>

6. Graphs

Subtopics

- Graph representation
- Adjacency lists
- BFS
- DFS
- Connected components
- Cycle detection
- Topological sort
- Shortest path
- Multi-source BFS
- Union Find
- Dijkstra's Algorithm

LeetCode

- Number of Islands

<https://leetcode.com/problems/number-of-islands/>

- Clone Graph

<https://leetcode.com/problems/clone-graph/>

- Course Schedule

<https://leetcode.com/problems/course-schedule/>

- Rotting Oranges
<https://leetcode.com/problems/rotting-oranges/>
- Pacific Atlantic Water Flow
<https://leetcode.com/problems/pacific-atlantic-water-flow/>
- Network Delay Time
<https://leetcode.com/problems/network-delay-time/>
- Redundant Connection
<https://leetcode.com/problems/redundant-connection/>
- Word Ladder
<https://leetcode.com/problems/word-ladder/>

7. Dynamic Programming and Recursion

Subtopics

- Base cases
- Recursive calls
- Backtracking
- Subsets
- Permutations
- Combinations
- Memoization
- Tabulation
- 1D DP
- 2D DP
- Knapsack
- Subsequences
- State transitions

LeetCode

- Climbing Stairs
<https://leetcode.com/problems/climbing-stairs/>
- House Robber
<https://leetcode.com/problems/house-robber/>
- Coin Change
<https://leetcode.com/problems/coin-change/>
- Longest Increasing Subsequence
<https://leetcode.com/problems/longest-increasing-subsequence/>
- Longest Common Subsequence
<https://leetcode.com/problems/longest-common-subsequence/>
- Unique Paths
<https://leetcode.com/problems/unique-paths/>
- Subsets
<https://leetcode.com/problems/subsets/>
- Permutations
<https://leetcode.com/problems/permutations/>
- Combination Sum
<https://leetcode.com/problems/combination-sum/>

8. Heaps

Subtopics

- Min heap
- Max heap
- Top K elements
- Kth largest / smallest

- Running median
- Merging sorted data
- Two-heap pattern

LeetCode

- Kth Largest Element in an Array
<https://leetcode.com/problems/kth-largest-element-in-an-array/>
- Top K Frequent Elements
<https://leetcode.com/problems/top-k-frequent-elements/>
- K Closest Points to Origin
<https://leetcode.com/problems/k-closest-points-to-origin/>
- Merge K Sorted Lists
<https://leetcode.com/problems/merge-k-sorted-lists/>
- Find Median from Data Stream
<https://leetcode.com/problems/find-median-from-data-stream/>
- Task Scheduler
<https://leetcode.com/problems/task-scheduler/>

9. Hashing

Subtopics

- HashMaps
- HashSets
- Frequency counting
- Duplicate detection
- Value-to-index mapping
- Grouping
- Prefix sum + hashmap

- Caching previously seen values

LeetCode

- Two Sum

<https://leetcode.com/problems/two-sum/>

- Contains Duplicate

<https://leetcode.com/problems/contains-duplicate/>

- Valid Anagram

<https://leetcode.com/problems/valid-anagram/>

- Group Anagrams

<https://leetcode.com/problems/group-anagrams/>

- Longest Consecutive Sequence

<https://leetcode.com/problems/longest-consecutive-sequence/>

- Subarray Sum Equals K

<https://leetcode.com/problems/subarray-sum-equals-k/>

10. Bit Manipulation

Subtopics

- AND
- OR
- XOR
- NOT
- Left shift
- Right shift
- Checking bits
- Setting bits
- Clearing bits

- XOR cancellation
- Counting set bits

LeetCode

- Single Number
<https://leetcode.com/problems/single-number/>
- Number of 1 Bits
<https://leetcode.com/problems/number-of-1-bits/>
- Counting Bits
<https://leetcode.com/problems/counting-bits/>
- Reverse Bits
<https://leetcode.com/problems/reverse-bits/>
- Missing Number
<https://leetcode.com/problems/missing-number/>
- Sum of Two Integers
<https://leetcode.com/problems/sum-of-two-integers/>

11. Sorting

Subtopics

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Counting Sort
- Bucket Sort
- Custom comparators

- Interval sorting

LeetCode

- Sort an Array
<https://leetcode.com/problems/sort-an-array/>
- Merge Intervals
<https://leetcode.com/problems/merge-intervals/>
- Insert Interval
<https://leetcode.com/problems/insert-interval/>
- Meeting Rooms II
<https://leetcode.com/problems/meeting-rooms-ii/>
- Largest Number
<https://leetcode.com/problems/largest-number/>

12. Searching

Subtopics

- Linear search
- Binary search
- Binary search on sorted arrays
- Binary search on answer
- Rotated arrays
- Lower bound
- Upper bound
- Matrix search

LeetCode

- Binary Search

<https://leetcode.com/problems/binary-search/>

- Search in Rotated Sorted Array

<https://leetcode.com/problems/search-in-rotated-sorted-array/>

- Find Minimum in Rotated Sorted Array

<https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/>

- Search a 2D Matrix

<https://leetcode.com/problems/search-a-2d-matrix/>

- Koko Eating Bananas

<https://leetcode.com/problems/koko-eating-bananas/>

- Find First and Last Position of Element in Sorted Array

<https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/>